

Experiment 16: Comparison of Classification Algorithms**Aim**

To implement and compare different **classification algorithms** and evaluate their performance.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the dataset.

Step 3: Separate the input features and target variable.

Step 4: Split the dataset into training and testing sets.

Step 5: Create different classification models.

Step 6: Train each model using the training data.

Step 7: Predict the classes for the test data.

Step 8: Calculate the accuracy of each model.

Step 9: Compare the accuracies and identify the best-performing classifier.

Program

```
# Experiment 16 - Comparison of Classification Algorithms

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

models = {
    "KNN": KNeighborsClassifier(n_neighbors=3),
    "Naive Bayes": GaussianNB(),
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Logistic Regression": LogisticRegression(max_iter=200)
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    print(name, "Accuracy:", accuracy)
```

Output

```
*** KNN Accuracy: 1.0  
    Naive Bayes Accuracy: 1.0  
    Decision Tree Accuracy: 1.0  
    Logistic Regression Accuracy: 1.0
```

Result

Thus, different **classification algorithms were successfully implemented and compared** based on their accuracy, and the performance of each classifier was evaluated.